Spencer Osborn

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Spencer Osborn
Splendero
Website

B. A. Sc (Mathematics and Engineering) Candidate
Queen's University

Education

Bachelor of Applied Science in Mathematics and Engineering (Communications and Computing) *2022-27*

Core concepts from both Computer Engineering and Applied Mathematics taught within the Faculty of Engineering.

Work Experience

Exiger *September 2026 - Current*

AI Data Engineer, Part-Time
Previously: AI Data Engineer Intern *June 2026 - September 2026*

- Designed, implemented, and launched an agentic workflow that automates due diligence report drafting for **200+ researchers**, reducing report writing time by **40%** and freeing **∼30,000 analyst hours** annually.
- Developed multi-stream web search and scraping pipelines using SerpApi, Firecrawl, and in-house tools to collect difficult-to-access due diligence data, navigating paywalls, CAPTCHAs, and site searches.
- Contributed to production **Kubernetes** deployments through Argo CD and implemented a Jira-based ticketing system, collaborating closely with domain experts to translate research workflows and requirements into reliable technical solutions.

Artificial Intelligence Software Engineering Intern *May 2025 - May 2026*

Novari Health

- Designed, developed, and implemented an intelligent document processing (IDP) into production, used by internal and external clients, using **C#**, **Azure**, and open-source frameworks to integrate AI features.
- Developed **NLP** solutions for IDP including: semantic matching, classification, and text extraction using LLMs.
- Designed and developed a document classification system using **OpenCV** and **Tesseract**, integrated into an internal library that supports large-scale document workflows, increasing classification accuracy by **50%**.
- Built and maintained internal metrics and monitoring tools for real-time performance analysis of IDP and Auto-Protocolling systems, with outage and underperformance alerting, leveraging data pipelines and **SQL** databases.

Software Engineering Intern *November 2024 - May 2025*

Limestone Analytics

- Built **Python** scripts to efficiently filter, sort, analyze, and update millions of rows of data, streamlining workflows and enabling timely, data-driven insights, using **Google** API integration.
- Developed a lightweight repository system to organize and manage data analysis tools, enabling better software reuse and version control for backend development workflows.

Design Team Experience

Co-Captain *September 2024 - Current*

aQuatonomous

- Leading administration, sponsorships, and operations while serving as the software lead, with a focus on computer–hardware integration to address computational efficiencies.
- Led a team of 6 building an Autonomous Surface Vessel for 2025 RoboBoat; delivered real-time object detection with **OpenCV**/**ROS2** on a Jetson Nano at **15 FPS** without impacting backend or external systems.

Personal Projects

Automatic Triage Agent

- Built an AI-driven medical triage backend using AWS Lambda, Bedrock, and EC2, enabling real-time patient prioritization through LLM-based symptom analysis and API-triggered workflows.
- Designed event-driven data pipelines with S3 and DynamoDB for asynchronous request handling.

TTC Visualizer

- Built a visualization webpage using **HTML** to display every TTC vehicle in Toronto using a public API.
- Created a backend that logs **1M+** vehicle locations in an **SQL** database and displays prior days' transit data.

Technical Skills and Interests

Languages: Java, Python, C/C++, C#, Go, HTML, JavaScript, PHP, LaTeX, SQL
Libraries/Frameworks: NumPy, OpenCV, Pandas, TensorFlow, Tesseract, React
Software: Azure, AWS, Docker, MS Office, GitHub, Linux, Kubernetes
Relevant Coursework: Real Analysis II, Algorithms, Control Systems, Probability II, Databases, Object-Oriented Programming, Complex Analysis, Linear Algebra II
Interests: Climbing, Volleyball, Cycling, Designing & Creating Rugs, Squash